

✓ Power Of Three

Question

Given an integer n , return true if it is a power of three. Otherwise, return false.

An integer n is a power of three, if there exists an integer x such that $n == 3^x$.

Example 1:

- Input: $n = 27$
- Output: true
- Explanation: $27 = 3^3$

Constraints

- $-2^{31} \leq n \leq 2^{31} - 1$
-

Solution

so for this question here i do it by using recursion..

Code

```
class Solution {
    public boolean helper(int num) {
        if (num == 0 ) {
            return false;
        }
        if (num == 1) {
            return true;
        } else {
            if (num % 3 != 0) {
                return false;
            }
            return helper(num / 3);
        }
    }

    public boolean isPowerOfThree(int n) {
```

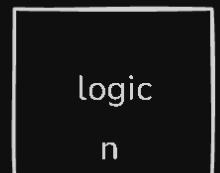
```
    if (n == 1) {  
        return true;  
    }  
    if (n % 3 != 0) {  
        return false;  
    } else {  
        return helper(n);  
    }  
}  
}
```

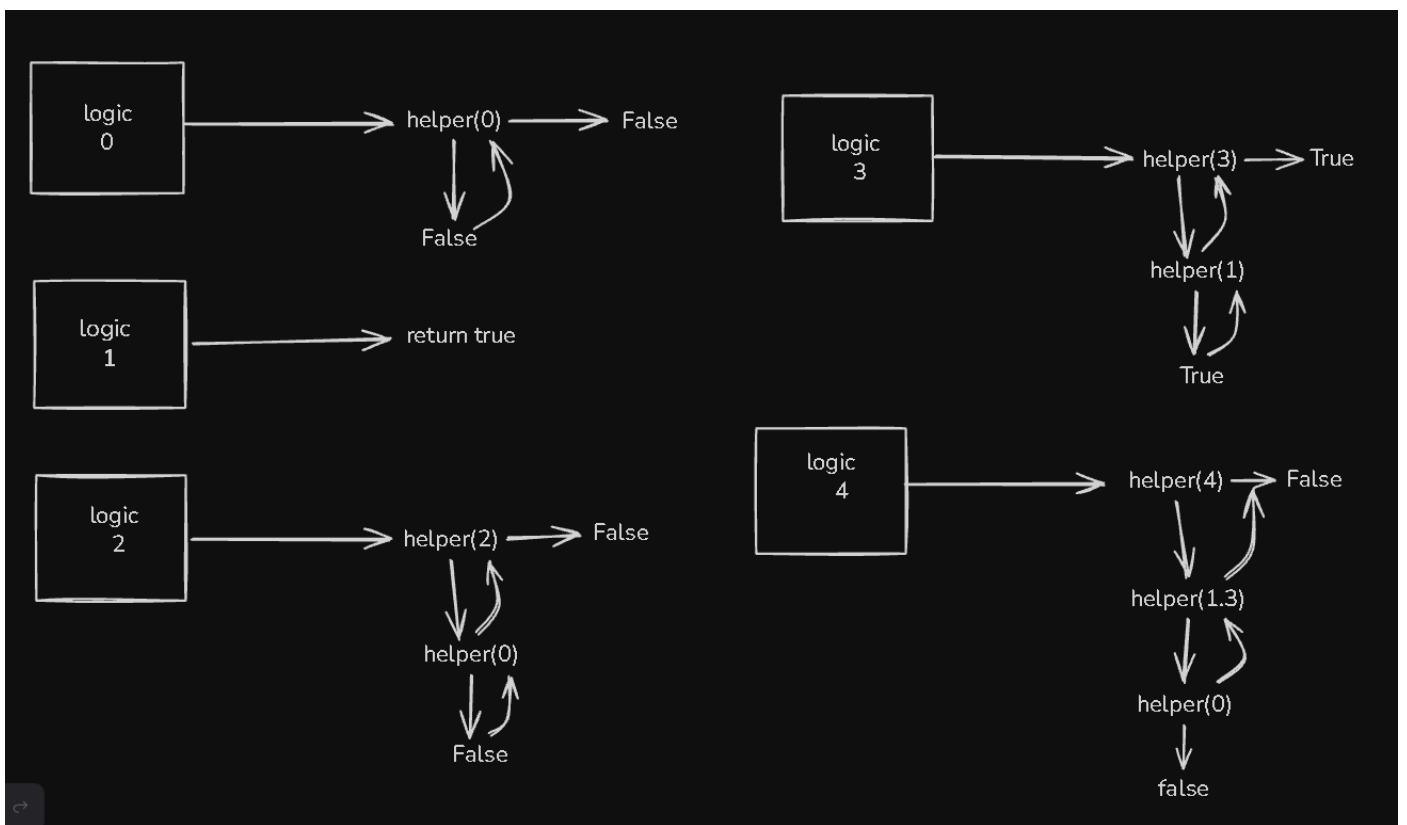
✓ dry run of above code

let run this code for 0 to 10 and examine the outputs.

input	output
0	False
1	True
2	False
3	True
4	False
5	False
6	False
7	False
8	False
9	True
10	False

```
isPowerOfThree(int n){  
    _____  
    _____  
    _____  
}
```





time complexity = $O(\log(n))$

space complexity = $O(1)$